

TRAVERSE CITY CHERRY CAPITAL, MI, USA

WMO: 726387

Lat: 44.741N

Lon: 85.583W

Elev: 618

StdP: 14.37

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	0.2	5.0	-6.3	4.0	1.6	-1.5	5.2	6.5	23.7	21.3	20.8	21.7	6.1	280	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.3	88.8	71.5	85.8	70.1	82.8	68.6	74.6	84.3	72.6	81.8	71.0	79.4	9.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.7	119.6	80.3	69.7	111.9	77.4	68.0	105.2	76.2	38.4	85.2	36.6	81.4	35.1	79.3	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.4	18.3	16.5	-6.6	94.0	7.3	2.7	-11.9	96.0	-16.2	97.6	-20.3	99.1	-25.6	101.1
			-7.1	77.8	7.2	2.2	-12.3	79.3	-16.5	80.6	-20.5	81.8	-25.8	83.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	22.7	23.5	31.9	43.1	54.9	65.4	70.1	68.7	62.0	50.1	38.5	28.9		
	DBStd	18.97	9.78	10.57	11.18	9.12	8.93	7.63	6.17	5.66	7.76	8.67	8.37	8.63		
	HDD50	3580	847	742	574	246	45	1	0	0	4	107	358	656		
	HDD65	7162	1312	1162	1028	657	336	86	22	26	148	470	795	1120		
	CDD50	2408	0	1	12	41	199	462	623	581	365	110	13	1		
	CDD65	513	0	0	1	1	25	97	180	142	59	8	0	0		
	CDH74	4567	0	0	14	34	313	958	1638	1099	450	61	0	0		
	CDH80	1330	0	0	3	5	91	316	511	285	109	10	0	0		
	WSAvg	6.8	7.8	7.8	7.4	7.6	6.5	5.7	5.9	5.6	5.9	6.8	7.5	7.5		
	PrecAvg	30.4	2.4	1.4	1.8	2.4	2.4	3.1	2.8	2.9	3.9	2.7	2.5	2.2		
Precipitation	PrecMax	39.2	10.7	4.1	7.3	5.4	5.7	9.2	9.8	7.8	9.2	7.1	4.7	4.2		
	PrecMin	22.2	1.0	0.3	0.2	0.9	0.3	0.4	0.3	0.5	0.4	0.9	0.4	0.9		
	PrecStd	4.4	1.7	0.9	1.3	1.1	1.2	2.0	1.8	1.7	2.2	1.5	1.0	0.9		
	PrecStd	4.4	1.7	0.9	1.3	1.1	1.2	2.0	1.8	1.7	2.2	1.5	1.0	0.9		
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	49.6	54.1	72.0	79.0	87.8	91.9	93.1	90.6	88.3	81.0	68.0	56.1		
		MCWB	46.2	46.2	58.2	60.6	69.0	74.0	74.5	72.4	70.9	67.0	56.2	52.7		
	2%	DB	43.4	46.9	60.8	72.0	82.1	87.6	88.5	86.4	82.7	74.4	60.3	48.0		
		MCWB	39.7	41.6	51.5	56.6	65.5	69.9	72.1	71.2	68.3	63.7	52.7	44.1		
	5%	DB	38.9	41.3	53.6	64.8	76.8	83.6	85.4	82.8	79.1	69.6	55.4	43.9		
		MCWB	36.2	37.1	46.3	51.4	62.1	68.3	70.7	69.4	66.9	60.2	48.8	40.1		
	10%	DB	36.2	37.3	47.2	58.9	71.4	79.7	82.3	80.2	75.4	64.0	51.5	40.0		
		MCWB	34.1	34.2	40.8	48.8	59.4	66.3	68.9	67.7	64.9	57.2	46.2	37.0		
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.1	48.4	60.4	63.7	71.7	75.9	78.0	76.0	73.4	70.1	58.5	53.1		
		MCDB	48.7	51.3	71.0	74.0	84.0	88.6	87.4	85.4	82.8	77.9	64.6	55.1		
	2%	WB	39.8	41.5	53.4	58.2	67.7	72.5	75.1	73.5	71.1	65.4	54.0	45.4		
		MCDB	42.5	46.1	59.1	68.2	77.8	84.0	84.7	82.0	78.4	72.0	59.0	47.5		
	5%	WB	36.4	37.6	46.2	53.5	64.3	70.4	72.9	71.7	69.1	61.2	50.2	40.4		
		MCDB	38.6	40.6	51.8	63.2	73.9	79.8	81.8	79.9	76.4	67.6	55.2	43.4		
	10%	WB	34.1	34.5	41.3	49.0	60.9	68.3	71.0	69.9	66.9	57.5	46.2	37.2		
		MCDB	36.1	37.4	47.0	58.1	69.7	76.9	79.4	77.0	74.1	63.7	50.7	39.8		
Mean Daily Temperature Range	5% DB	MDBR	11.4	14.0	16.6	19.9	21.8	20.9	20.3	19.1	19.0	16.2	12.5	10.1		
		MCDBR	13.8	17.3	24.1	30.8	28.1	25.1	23.3	22.3	22.8	23.2	19.4	15.1		
		MCWBR	11.9	13.3	16.4	17.9	14.8	11.9	10.8	10.5	11.4	13.6	13.7	12.8		
	5% WB	MCDBR	12.9	15.9	21.8	27.5	24.8	21.1	19.3	18.2	19.1	19.9	17.6	14.0		
		MCWBR	11.5	12.8	16.1	18.0	14.8	11.6	10.2	9.8	11.0	14.0	13.8	12.7		
Clear-Sky Solar Irradiance	taub		0.284	0.295	0.321	0.355	0.384	0.402	0.397	0.387	0.372	0.341	0.308	0.283		
	taud		2.380	2.327	2.391	2.349	2.332	2.308	2.341	2.364	2.396	2.479	2.506	2.438		
	Ebn at Noon		266	283	290	287	281	275	275	274	269	264	256	255		
	Edh at Noon		24	30	33	37	39	40	38	36	32	26	21	21		
All-Sky Solar Radiation	RadAvg		379	672	1126	1443	1759	1920	1936	1648	1286	758	420	281		
	RadStd		36	70	126	141	154	134	89	108	119	92	60	31		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page